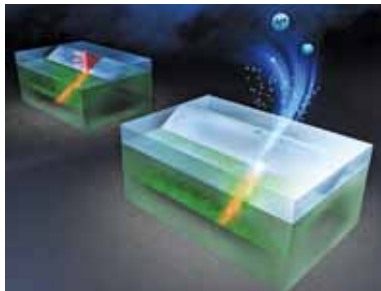




machine interface device. The participants were asked to control the direction of the wheelchair by thinking about moving their body parts. They observed that this could be done not only through machine learning of the device but also learning in the brain of the participants. Over the course of training, the brain-machine interface device team saw significant improvement in accuracy of over 95%.

Proton implantation prevents single Shockley stacking

Silicon carbide (SiC) is a promising alternative to silicon for power electronics devices, but it has low reliability as it eventually degrades. Over a period of time, small dislocations take place in the crystal structure, which eventually



Silicon carbide (SiC) is a promising alternative to silicon for power electronic devices, but has low reliability for undergoing bipolar degradation. Now, researchers from Japan have proposed a method that can suppress bipolar degradation in SiC via proton implantation into the substrate using a particle accelerator (Credit: Masashi Kato from Nagoya Institute of Technology)

turn into large defects. These defects are called single Shockley stacking faults. Researchers from the Nagoya Institute of Technology have developed a solution called proton implantation which prevents degradation. Proton implantation involves injecting hydrogen ions into the substrate using a particle accelerator. The idea is to prevent the formation

of single Shockley stacking faults by pinning down partial dislocations in the crystal, one of the effects of introducing proton impurities. However, proton implantation itself can damage the 4H-SiC substrate, due to which high-temperature annealing is used as an additional processing step to repair this damage. Results showed that the diodes that had undergone proton implantation performed as well as regular ones but without signs of bipolar degradation.

Nerve-operated prosthetics show the way for better control

Researchers at University of New South Wales, before the pandemic, had theoretically proved that sensors developed using liquid crystal and integrated optics technology—dubbed ‘optrodes’—can detect nerve impulses in a living animal body. These optrodes use light instead of electricity to measure neural activity. Recently they demonstrated that optrodes could be used to accurately measure neural impulses as they moved along a nerve fibre in a living animal. They connected an optrode to the sciatic nerve of



Optrodes have the potential to make future prosthetics as functional as the limb they replace (Credit: <https://scitechdaily.com>)

an anesthetised animal. The nerve was then stimulated with a small current and the neural signals were recorded with the optrode. They demonstrated that nerve impulses—which are

relatively weak and measured in microvolts—can be registered by optrode technology. In future, a chip with thousands of optical connections could connect to your brain, or someplace in the arm before the nerve bundle separates to connect to a prosthetic hand, and the hand could function much the same way as a biological one.

Better resolution of 3D printing now made possible

The problem with current 3D printing technology is that it does not have enough resolution for the print. Researchers from Stanford University have developed a solution to this problem. Instead of writing metals directly, they 3D print a hydrogel and use it as a scaffold for metal-containing liquid precursors. Hydrogels are made from flexible polymer chains which do not dissolve in water. When these hydrogels are exposed to light from low-powered ultraviolet lamps, the polymer chains cross-link resulting in hardening of the hydrogel. Researchers then infused the 3D printed hydrogel scaffolds with metal salts dissolved in water, causing the metal ions to infiltrate the hydrogel—not just coat its surface. Then, in reaction portion of the process, the researchers burn away the hydrogel portion of the structure in a furnace that reaches 700 to 1100 degrees Celsius, depending on the material.

Better technique found for large optical neural networks

Researchers from MIT have developed a way to effectively scale an optical neural network for analogue devices. By adding a tiny hardware component to the optical switches that form the network’s architecture they can reduce even the uncorrectable errors that would otherwise accumulate in the device. Researchers have developed a new type of MZI, which consists of an additional beam splitter to the end of the device, naming it ‘3-MZI’ because it has three beam splitters instead of two. Due to the way this additional beam splitter mixes the light, it becomes much easier for an MZI to reach the setting it needs to send all light through its bottom port. The team found that it can eliminate much of the uncorrectable error that hampers accuracy. And as the optical neural network becomes larger, the amount of error in the device actually drops—the opposite of what happens in a device with standard MZIs.